

Cell Design Specification — t4_r3

Case t4_r3 | 2026-08-26 | generated by bda (headless) | PDF release

Basic specification

■ ■	■	■ ■
■■■■■■■	NMC■■■■ / ■■■■■■ (OKane2022 ■■■■■■, LGM50 ■■■)	param_dump_okane.json (■■■■) [sourced]
■■■■■	5.0 Ah (■■■■■ 5.0383 Ah @1C 25C DFN)	param_dump_okane.json / r7_V18_1c_25C_dfn.json:capacity_ah
■■■■■	2.5 - 4.2 V	param_dump_okane.json
■■■■■	■■ 65 mm (■) x 1580 mm (■, ■■) x 182.8 um (■■■■■: ■■ 75.6 + ■■ 85.2 + ■■ 8 + Al 8 + Cu 6 um)	param_dump_okane.json + r7_V18_energy.json:thickness_m
■■■■■	Not provided (■■■■■■■■■■)	honest
■■■■■■■	■■■■■■■: sigma=2.2 S/m, D=6.0e-10 m2/s, t+=0.5 (■■■■■■■■■■ — ■■■■■■ Stage 3 ■■, ■■ Stage 2 ■■■■■)	r7_V18_params.json [estimated, ■■■■■]
■■■■■■■	0.5 (■■■■■ 0.2594)	■■

Electrodes and separator

■	■■ um	■■■	■■/■■ kg/m3	■■
■■■■■■	75.6	0.335	3262	param_dump_okane.json
■■■■■■	85.2	0.25	1657	param_dump_okane.json
■■	8 (■■ 12)	0.47	397	r7_V18_params.json + param_dump_okane.json
■■■■■■	8 (■■ 16)	--	Al 2700	r7_V18_params.json + param_dump_okane.json
■■■■■■	6 (■■ 12)	--	Cu 8960	r7_V18_params.json + param_dump_okane.json
N/P	1.127 (■■■ 85.2/75.6; ■■■■■■■■■■ — ■■■■■■■■■■; ■■■■■■■■■■, ■■■■■■■■■■, ■■2■■■■■ N/P~1.18 ■■■■)	--	param_dump_okane.json [inferred] + log.jsonl R2	

Process design parameters

■ ■	■	■ ■	■ ■
■ ■ ■ ■ ■ ■	163.99 g/m2	■ ■ x (1-■ ■ ■ ■) x ■ ■	[inferred ■ ■ ■ ■ ■]

Item	Value	Unit	Notes
Weight	105.88	g/m2	[inferred Weight]
Density	2.169	g/cm3	[inferred Density]
Volume	1.243	g/cm3	[inferred Volume]
Mass	6.210	g	[estimated, Weight]
Capacity	0.1C CC @ 4.2V, 25C, 2		[design recommended]

Mass breakdown (g/cell)

Item	Value g	Unit	Notes
Weight	16.842	layer_kg_m2 x Weight (0.1027 m2)	r7_V18_energy.json:layer_kg_m2 [sourced]
Weight	10.874		[sourced]
Al Weight (8um)	2.218		[sourced]
Cu Weight (6um)	5.521		[sourced]
Weight (8um)	0.173	Weight (1-Weight) Weight	[sourced]
Weight (Weight)	35.629	Weight (calc-energy Weight)	[sourced]
Weight (Weight)	6.210	Weight x 1.2 g/cm3	[estimated]
Weight (Weight)	41.838	Weight	[inferred]

Performance verification vs entry-0 criteria

Item	Value	Unit (entry-0, Unit)	Notes	Notes
1C Weight (25C, DFN)	5.0383 Ah	Weight 5.0 Ah (Weight)	PASS Weight	r7_V18_1c_25C_df n.json:capacity_ah
-20C Weight	97.97%	min 95	PASS	bridge/r7_V18_lowT _retention.json [inferred]
Weight	514.8 Wh/kg	min 327.18	PASS	r7_V18_energy.json [sourced]
Weight	977.0 Wh/L	min 880	PASS	r7_V18_energy.json [sourced]
4C Weight	330.45 K (57.30C)	max 333.15 K	PASS	r7_V18_4C_45C_df n.json:T_max_K
4C Weight	Weight +0.0155 V	plated = false (Weight<0)	PASS (Weight, Weight)	r7_V18_4C_45C_df n.json:anode_potential_v

Design notes

Notes

[illegible]